Ruizhe Xu

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Education

2026 – Now **The University of Hong Kong**, Hong Kong, China.
Postdoctoral Fellows in Department of Mathematics.
Supervised by *Prof. Zhigang Bao*.

2021 – 2026 **Zhejiang University**, Hangzhou, Zhejiang, China.
Ph.D. Student in Probability Theory and Mathematical Statistics.
Supervised by *Prof. Zhonggen Su*.

2025 – 2026 **University of Pennsylvania**, Philadelphia, Pennsylvania, US.
Visiting Ph.D. Student in Department of Statistics and Data Science.
Supervised by *Prof. Jiaoyang Huang*.

2017 – 2021 **Zhejiang University**, Hangzhou, Zhejiang, China.
Bachelor in Mathematics and Applied Mathematics.

Research Interests

Probability theory, especially random matrix, random graphs, stochastic homogenization, and random walk in random environment.

Publications/Preprints

[1] **Coupling between Brownian motion and random walks on the infinite percolation cluster**, with Chenlin Gu and Zhonggen Su, *Probability Theory and Related Fields* 196, 443–485 (2026).

[2] **Quantitative Einstein relation for reversible diffusions in a random environment**, with Ahmed Bou-Rabee, *arXiv* https://doi.org/10.48550/arXiv.2605.26082.

[3] **Non-symmetric vector Dyson equations**, with Jiaoyang Huang and Zhonggen Su, *arXiv*, https://doi.org/10.48550/arXiv.2607.16333.

[4] **Local Law and Eigenvector Delocalization for Random Lifts**, with Jiaoyang Huang, *arXiv*, https://doi.org/10.48550/arXiv.2610.37463.

Talks

10/2026 **14th National Conference on Probability and Statistics**, Jiangxi, China.

11/2025 **24th Northeast Probability Seminar**, the CUNY Graduate Center, New York, US.

05/2025 **Yangtze River Delta Probability & Statistics Seminar**, Shanghai University of Fiance and Economics, Shanghai, China.

Awards

06/2026 **Outstanding Graduate**, Zhejiang University.

10/2025 **National Scholarship**, Zhejiang University.

09/2022 **Outstanding Graduate Student**, Zhejiang University.

09/2021 **Freshman Scholarship for Doctoral Candidate**, Zhejiang University.

06/2021 **Outstanding Undergraduate**, Zhejiang University.

09/2019 **Jin Ming Education Scholarship**, Zhejiang University.

09/2018 **Academic scholarship**, Zhejiang University.

Teaching

Fall 2021 & Fall 2022 **Teaching Assistant**, Probability Theory, Zhejiang University. for undergraduates students

Fall 2022 & Fall 2023 **Teaching Assistant**, Advanced Probability Theory, Zhejiang University. for graduates students

Spring 2024 **Teaching Assistant**, Stochastic Processes, Zhejiang University. for graduates students